

Quantum Rope Perturbation Theory

The One-Loop Fluctuation Determinant — and Why It Falsifies the One-Loop Mass Mechanism

Mark Palmer · with computational collaboration by Claude (Anthropic)

Charlotte, NC · palmer100@gmail.com

Abstract

This paper reports the semiclassical (one-loop) perturbation theory of fluctuations around a rope soliton, and it reports a **NEGATIVE** result, plainly. The quantity that would control a fluctuation-generated mass is the regularised log-determinant of the second-variation operator, $\ln[\det'(-\nabla^2 + V)/\det(-\nabla^2)]$. Computed for the physical (positive, localised) second-variation potential, this ratio is an $O(1)$ number with **NO** negative modes — the solver returns -1.29 for a representative well. The electron scale, by contrast, would require a log-determinant of about 108 . The one-loop fluctuation determinant is therefore roughly two orders of magnitude too small, and of the wrong character, to generate the lepton mass scale. The one-loop mass mechanism is falsified by its own computation. We report this as a finding. All numbers are outputs of the rope_solver spectrum computation in this package (`log_det_ratio`, `poschl_teller_bound_states`, `required_log_det_for_electron`), not inserted constants.

The negative result, stated plainly

The one-loop fluctuation log-determinant around a rope soliton is $O(1)$ (computed ≈ -1.29) with no negative modes. The electron scale requires ≈ 108 . The one-loop determinant is $\sim 80\times$ too small and lacks the instability (negative mode) a mass-generating mechanism would need.

CONCLUSION: quantum fluctuations at one loop do **NOT** generate the lepton mass scale in the rope model. This is a falsification of the one-loop mass mechanism, kept as a finding, not removed.

1. One-Loop Perturbation Theory Around a Soliton

A classical rope soliton contributes to quantum amplitudes with a weight set, at one loop, by the determinant of the operator governing small fluctuations around it — the second variation of the action, $M = -\nabla^2 + V$, where V is the (positive, localised) second-variation potential. The physically meaningful, regularised quantity is the ratio to the free determinant,

$$\Delta = \ln[\det'(-\nabla^2 + V) / \det(-\nabla^2)] \quad (1.1)$$

computed from the low-lying spectra of the two operators on a common three-dimensional grid. If fluctuations were to generate a large mass scale, Δ would need to be large (and typically the spectrum would carry a negative mode signalling an instability).

2. The Computation and Its Result

Evaluated for the physical second-variation potential, the solver returns a small, negative, $O(1)$ value with no negative modes:

Quantity	Computed	What a mass mechanism needs
One-loop log-det Δ	≈ -1.29 ($O(1)$)	large (≈ 108 for the electron)
Negative modes	0	≥ 1 (instability) typically required
Ratio to electron requirement	$\sim 1/80$	order unity

The gap is not marginal. The one-loop determinant is about two orders of magnitude below the value the electron scale requires, and it lacks the negative mode that a genuine mass-generating instability would exhibit. No reasonable adjustment within one-loop perturbation theory closes a factor of eighty.

3. What This Falsifies, and What Remains

The result is specific and worth stating precisely.

3.1 Falsified

- The one-loop fluctuation mass mechanism: the hypothesis that quantum fluctuations around a rope soliton, at one loop, generate the lepton mass scale. The computed determinant is far too small and structurally wrong (no negative mode). This is a clean falsification by the programme's own computation.

3.2 What remains open

- Whether some non-perturbative or higher-order mechanism could supply the required scale is not addressed here and is not claimed. The negative result is specific to one-loop perturbation theory.
- The electron log-determinant requirement (≈ 108) remains, as reported in the knot-fluctuation paper, a TARGET that no computed rope spectrum has been shown to meet. This paper closes off the one-loop route to that target.

Why report it. A natural and attractive idea — that the lepton mass emerges from quantum fluctuations around the knot — is tested directly and fails at one loop, by a large and unambiguous margin. Recording that failure prevents the corpus from resting on a mechanism its own computation rules out, and narrows the search: whatever sets the lepton mass in the rope model, it is not the one-loop fluctuation determinant.

4. Status of Each Result

Result	Status
One-loop log-det $\Delta \approx -1.29$, $O(1)$, no negative modes	Derived — direct spectral computation
Electron requires $\Delta \approx 108$	Modeled — target (see knot-fluctuation paper)
One-loop fluctuation mass mechanism	Failed — falsified by $\sim 80\times$ gap; kept as finding
Non-perturbative mass mechanism	Open — not addressed, not claimed

Conclusion

Quantum rope perturbation theory, carried to one loop, returns a negative result: the fluctuation log-determinant around a rope soliton is an $O(1)$ quantity (computed ≈ -1.29) with no negative modes, while the electron scale would require roughly 108. The one-loop mechanism for generating the lepton mass from fluctuations is thereby falsified by the model's own computation, by about two orders of magnitude. In keeping with the programme's practice, the failure is reported as a finding and retained: it rules out an attractive but incorrect mechanism, and it sharpens the open question of what, if anything, sets the lepton mass scale in the rope model — which is not, the computation shows, one-loop fluctuations.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **QB-004 [Failed]:** One-loop fluctuation mass mechanism

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.